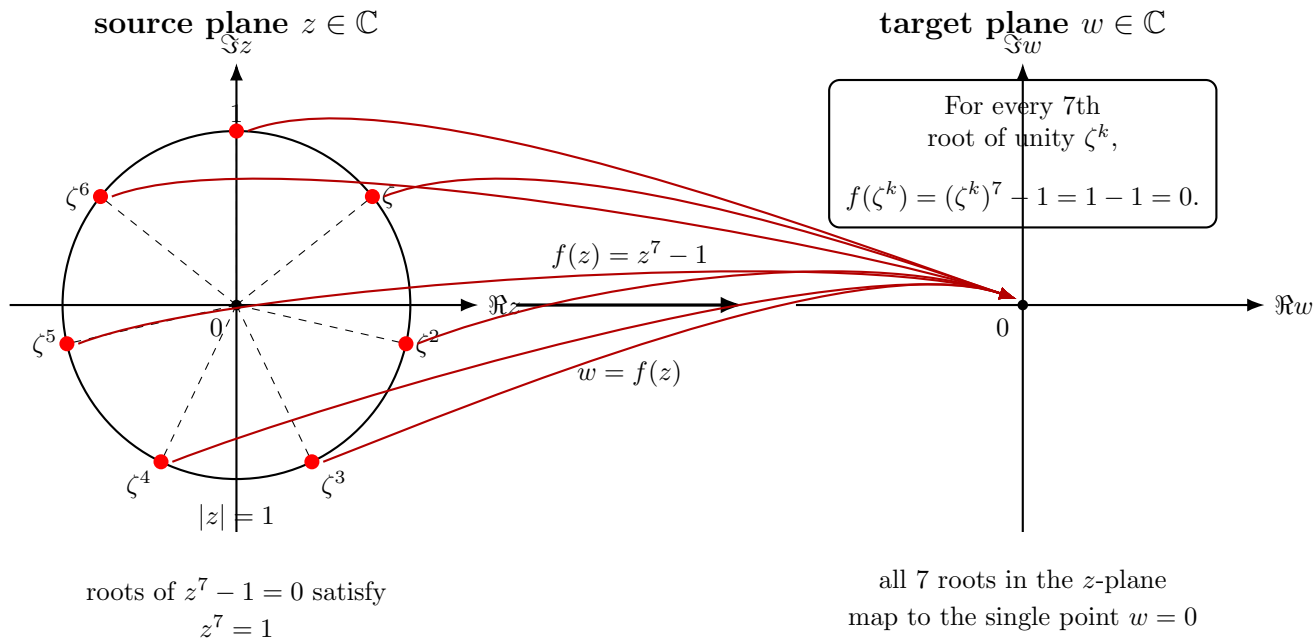


Visualization of $f(z) = z^7 - 1 : \mathbb{C} \rightarrow \mathbb{C}$



The equation $z^7 - 1 = 0$ means: find those points in the z -plane that map to 0 in the w -plane.

So the 7th roots of unity are exactly the preimages of 0 under $f(z) = z^7 - 1$.